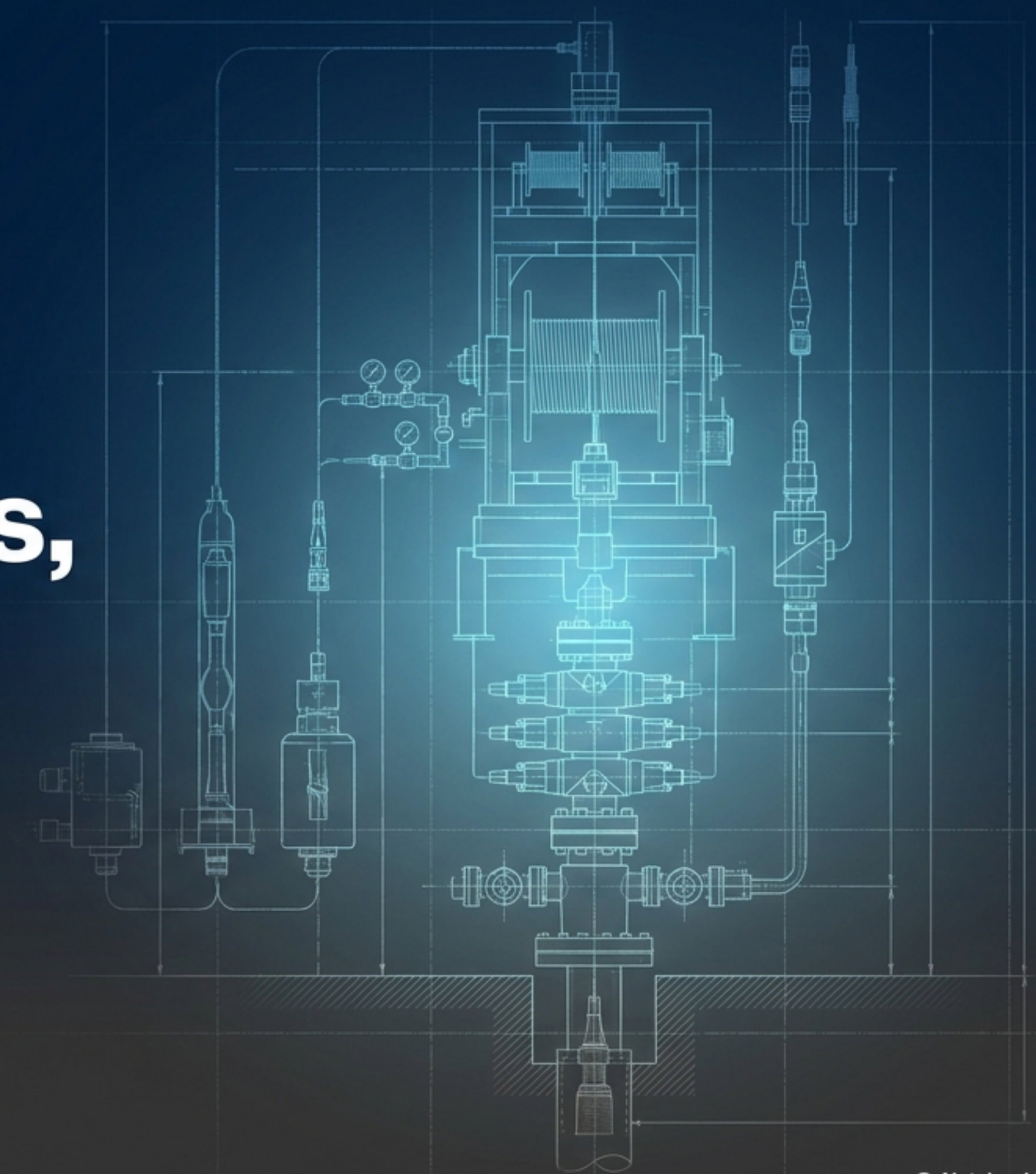


The Deepwater Blueprint.

The Wireline Nervous System: Diagnostics, Risks, and Reality

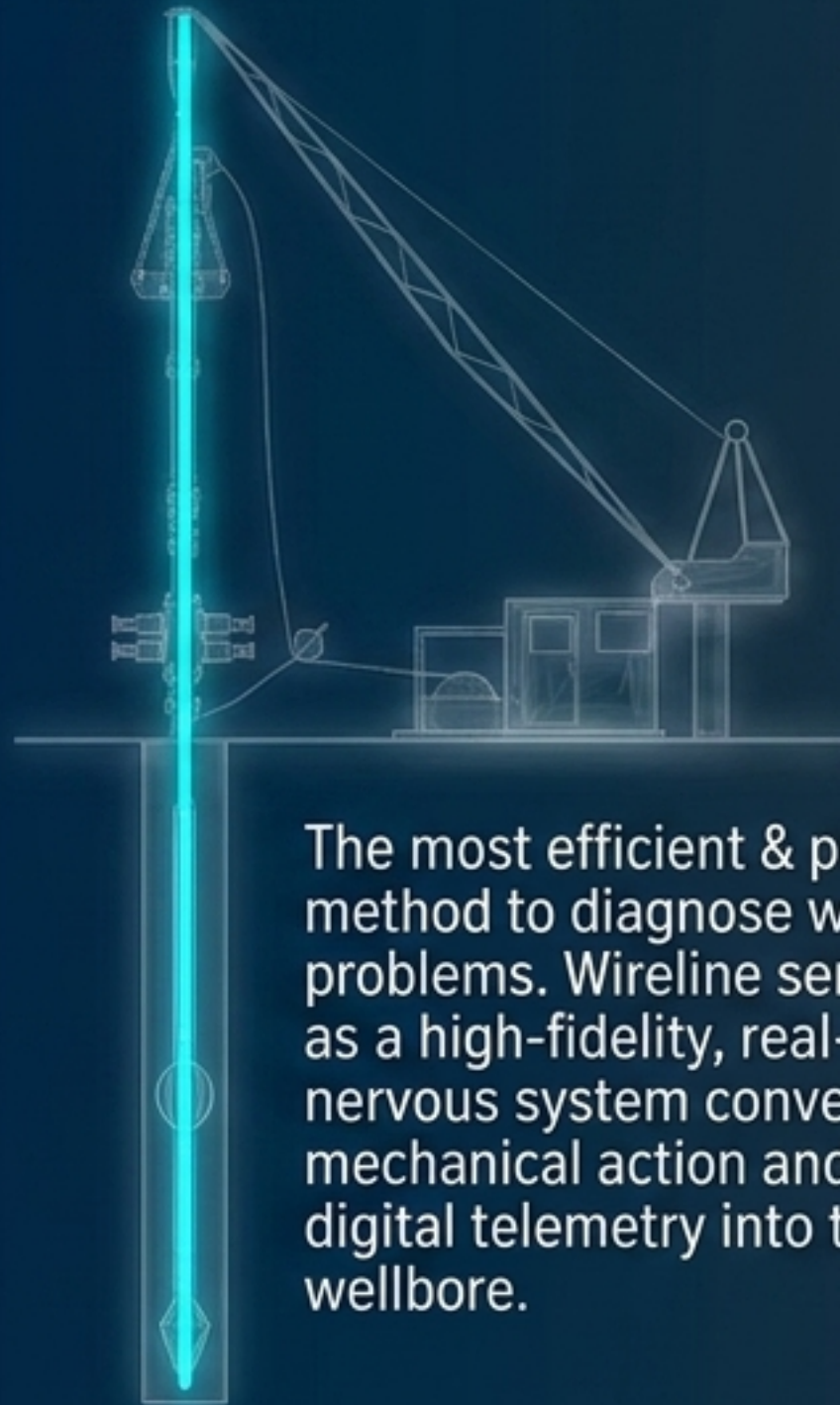
A Metacognitive Masterclass
in Well Interventions

Prepared by Eng. Sajjad k. Hano



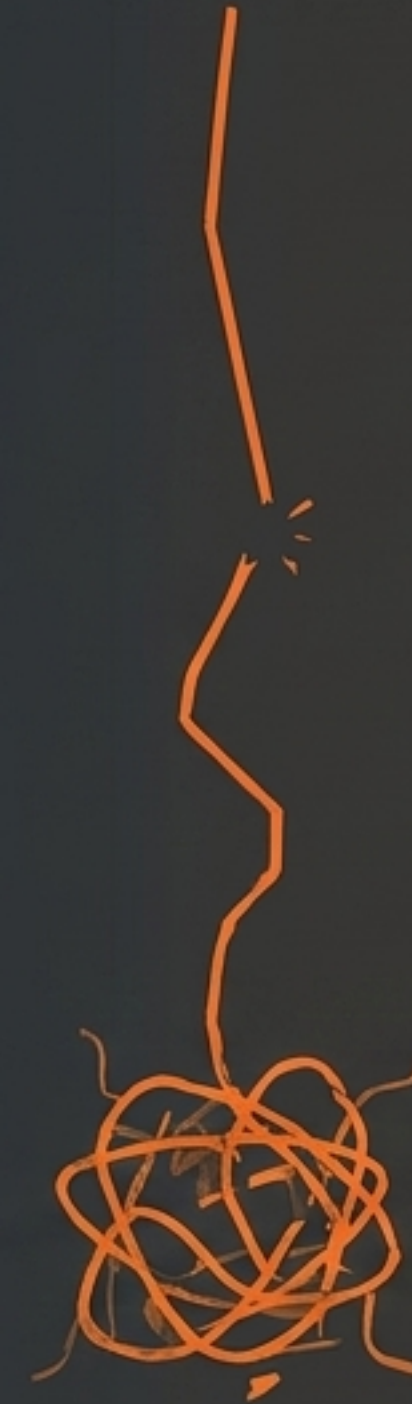
The Double-Edged Diagnostic

The Nervous System



The most efficient & practical method to diagnose well problems. Wireline serves as a high-fidelity, real-time nervous system conveying mechanical action and digital telemetry into the wellbore.

The Catastrophic Vulnerability



Easily junk wells if not properly managed. A junk-time-enter line system, a splashing problem for diagnostic tool into end emit a junked well.

Easily junk wells if not properly managed. A single-point failure architecture where poor planning transforms a diagnostic tool into permanent downhole debris.

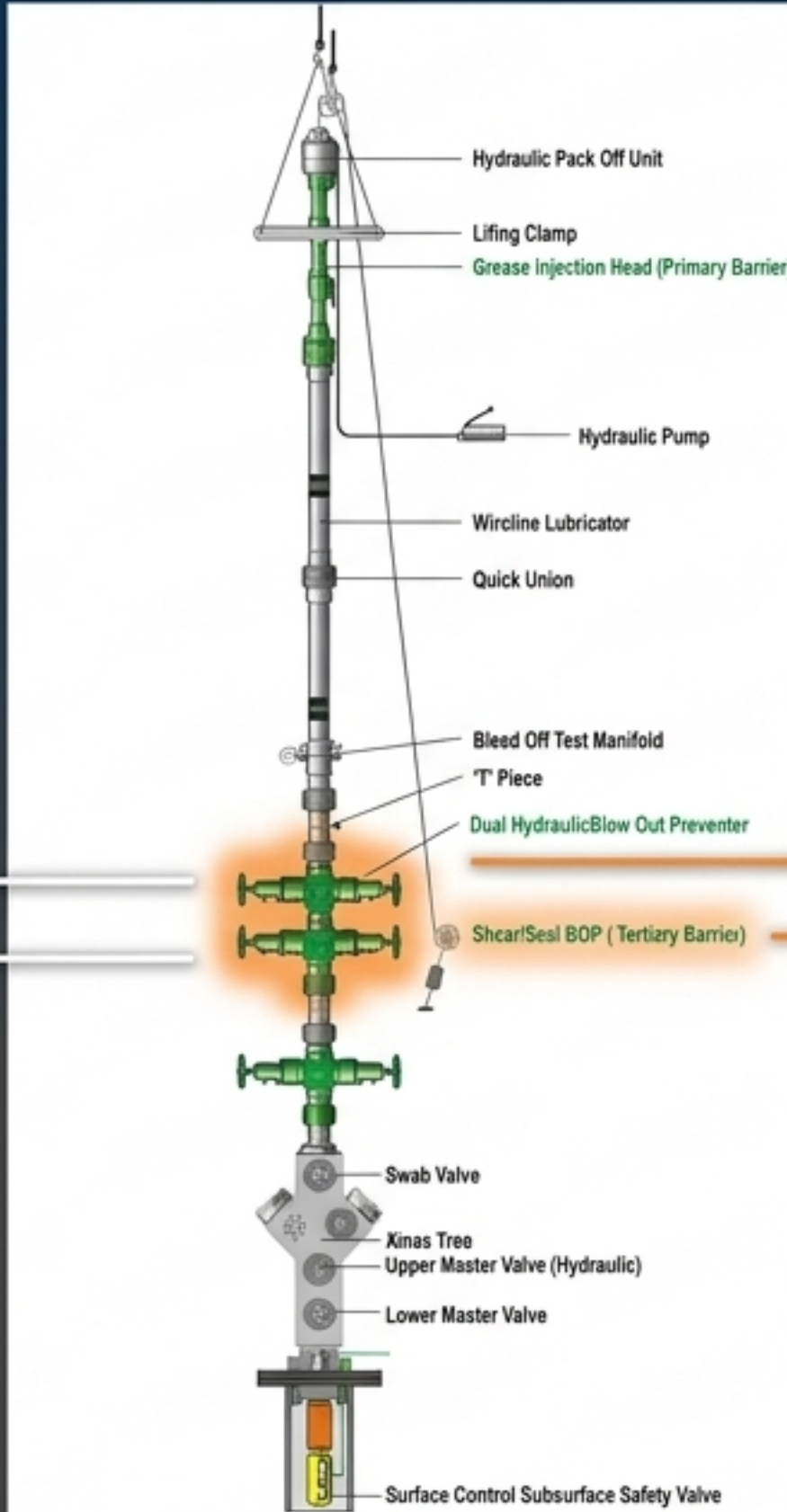
The "BUT..." of the Tertiary Barrier

The Position

Flanged immediately above the Xmas tree.

The Capability

Cuts up to 8-10 conductive wires. Holds pressure from below.



The Hidden Flaw

CANNOT cut tool strings or running tools.

The Aftermath

Once cut, wire falls below Xmas tree valves, permitting UMW closure.

Tactical Override

If the Upper Master Valve (UMV) has wire-cutting ability, use it instead. In an emergency, a standard gate valve cuts one strand of 7/32" wire, but a swab valve is preferred because it is easily replaced.

Fighting the Invisible Enemies: Chemistry & Static



The Chemical Shield

Pressure Testing Medium: Strict 60:40 ratio (Fresh Water : Glycol).

Hidden Implication: This precise ratio is not just for pressure—it specifically prevents Emulsion formation in oil wells and Hydrates when discharged into the well.

Minimum PT is Max SITHP, but 1.2x is recommended.

The Physics of Bleed-Off

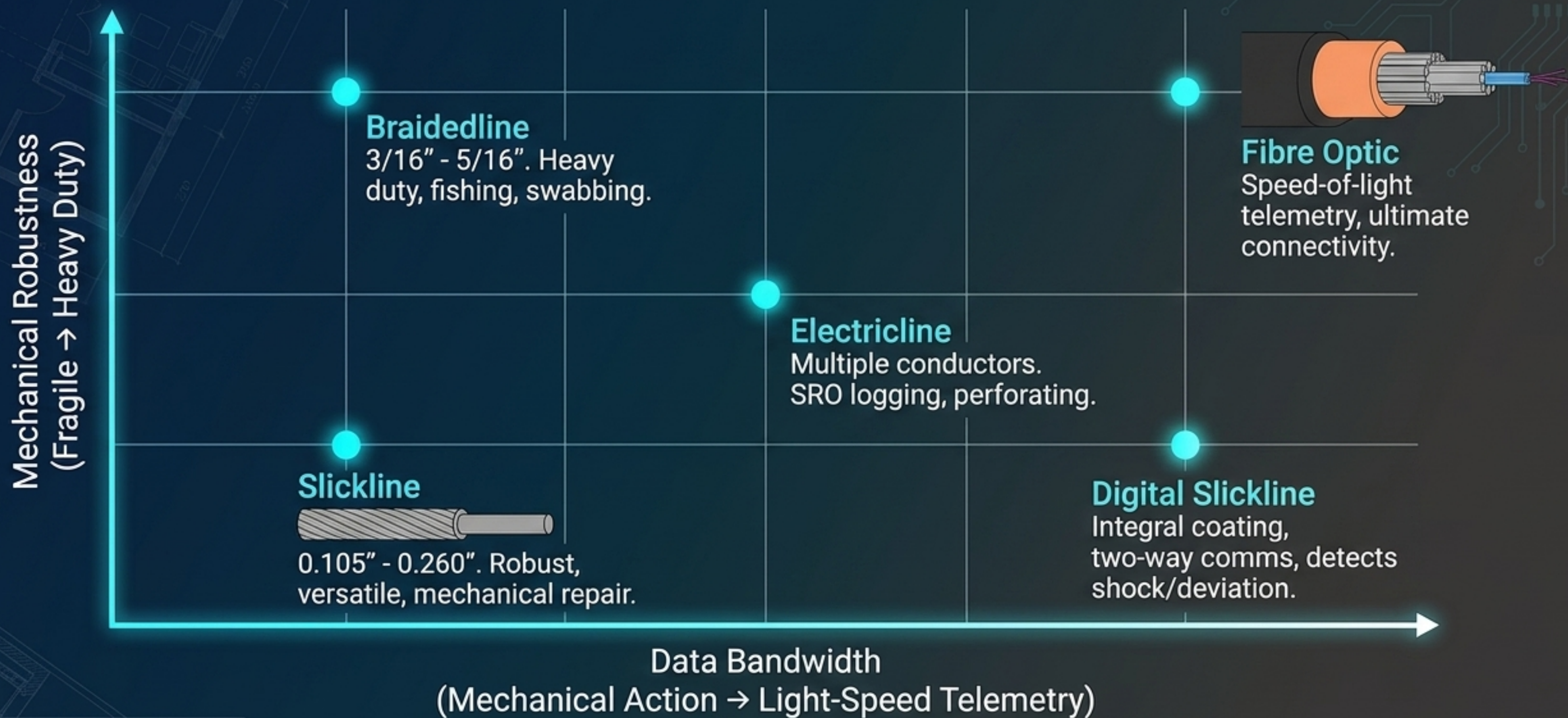
PATIENCE is the virtue.
NO ACCELERATION permitted.

Hidden Implication: Rushing de-pressurization invites static electricity ignition. Equipment must be heavily earthed.



Prepared by
Eng. Sajjad k. Hano

The Evolutionary Matrix of Conveyance



Cable Architecture & Tensile Dynamics

Conventional Slickline

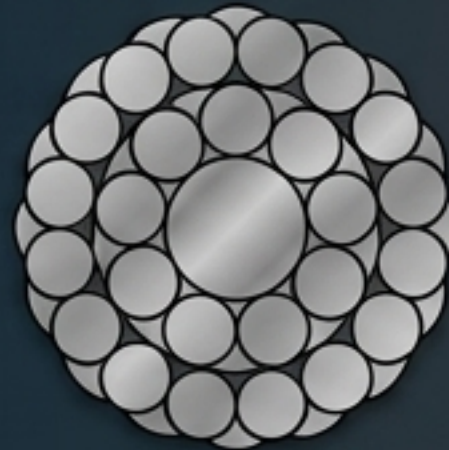


Profile: Single strand.

OD Range: 0.105" to 0.260".

Rule: Selection strictly depends on the severity of the operation load.

Conventional Braided



Profile: 16 strands.

Application: High-tension retrieval, conventional swabbing.

Dyform Braided



Profile: 19 strands. Low twist tendency.

The Variable Advantage:
Smooth external surface.
Contains more steel in the same diameter, resulting in a 20% increase in breaking load.

The Triangulation of Well Intervention

The Field Supervisor (Mechanical)

"Digital sensors are great, but **steel doesn't** lie. A blind box and a lead impression block give me ground truth. You cannot intervene in a well until you **physically drift it.**"

The Well Control Engineer (Risk)

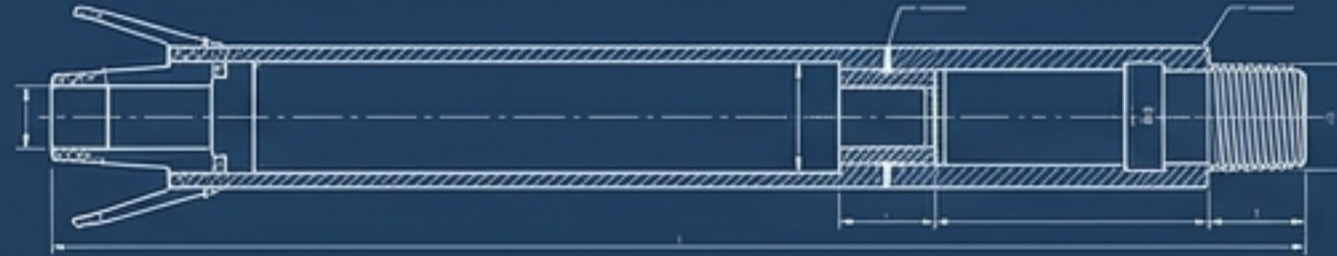
"Every cable is just a potential leak path. I don't care about your data speed; I care that the Shear/Seal won't cut a tool string, and that a rushed bleed-off will trigger a static ignition."

The Data Specialist (Digital)

"We are flying blind with conventional mechanical jarring. **Digital Slickline** gives us two-way communication, shock detection, and real-time **deviation** monitoring. Telemetry is the ultimate risk mitigation."

The Diagnostic Funnel: Slickline Drift Runs

1



Tubing End Locator & Sample Bailer

Function: Evaluates Hold-Up Depth (HUD) and tubing end.

2



Gauge Cutter

Function: Establishes tubing is clear. Scrapes clean paraffin, wax, and debris from the inside wall.

3



Blind Box

Function: Establishes exact liquid levels in the tubing.

4



Lead Impression Block (LIB)

Function: Gets a physical "picture" of a wireline fish before committing to a retrieval method.



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The Architecture of Failure: What If We Do The Opposite?

Skip the Drift Run

Action: Running high-value E-Line tools without a prior Gauge Cutter run.



Catastrophe: Mechanical Destruction

Tools hit unknown Hold-Up Depths, are destroyed by unscraped paraffin, or trigger catastrophic fishing jobs.

Accelerate De-Pressurization

Action: Bleeding off pressure rapidly to save rig time.



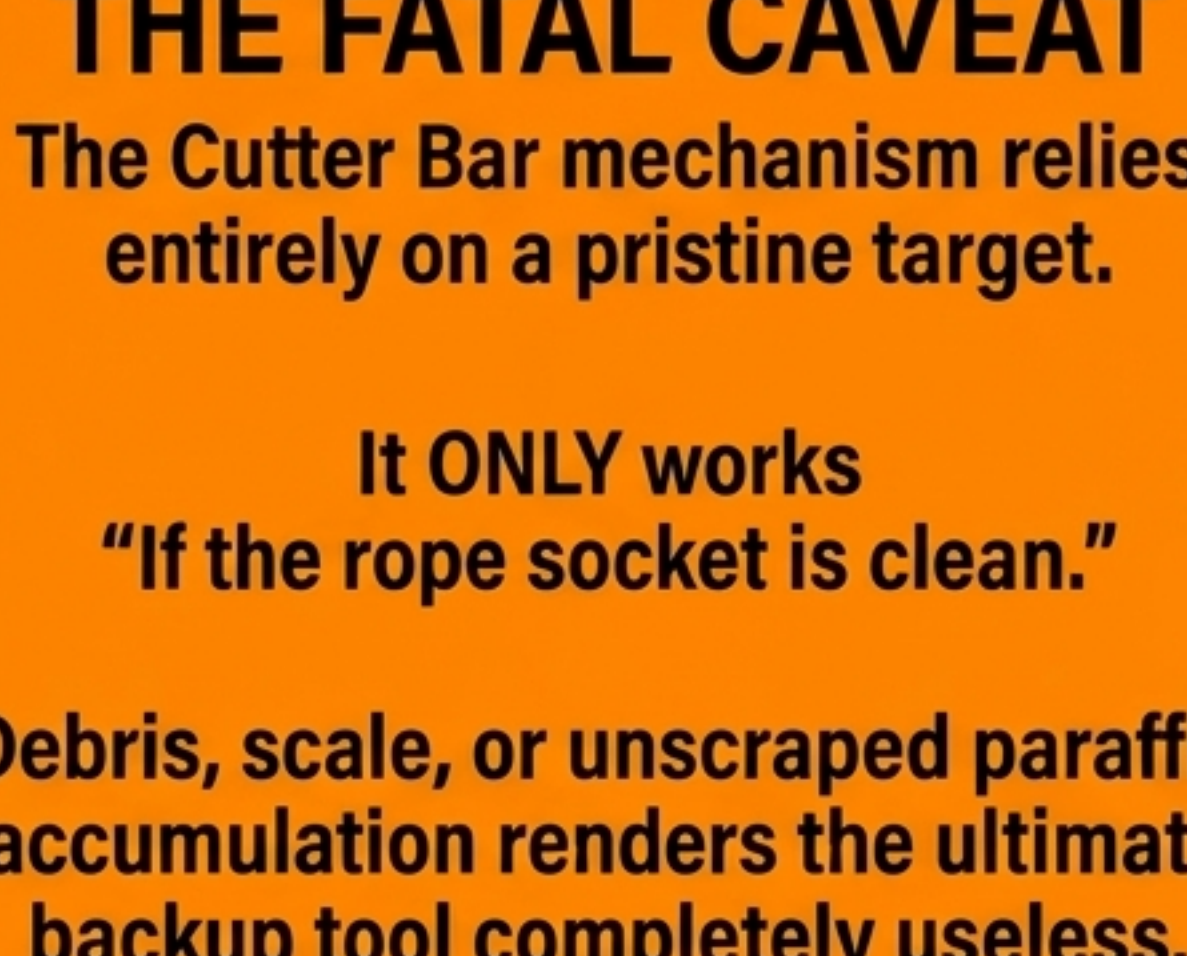
Catastrophe: Thermodynamic & Static Ignition

Uncontrolled thermodynamic shifts cause instant Hydrate formation, while high-velocity gas friction triggers static electricity ignition.

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Cognitive Load & The 5 Pillars of Preparation





THE FATAL CAVEAT

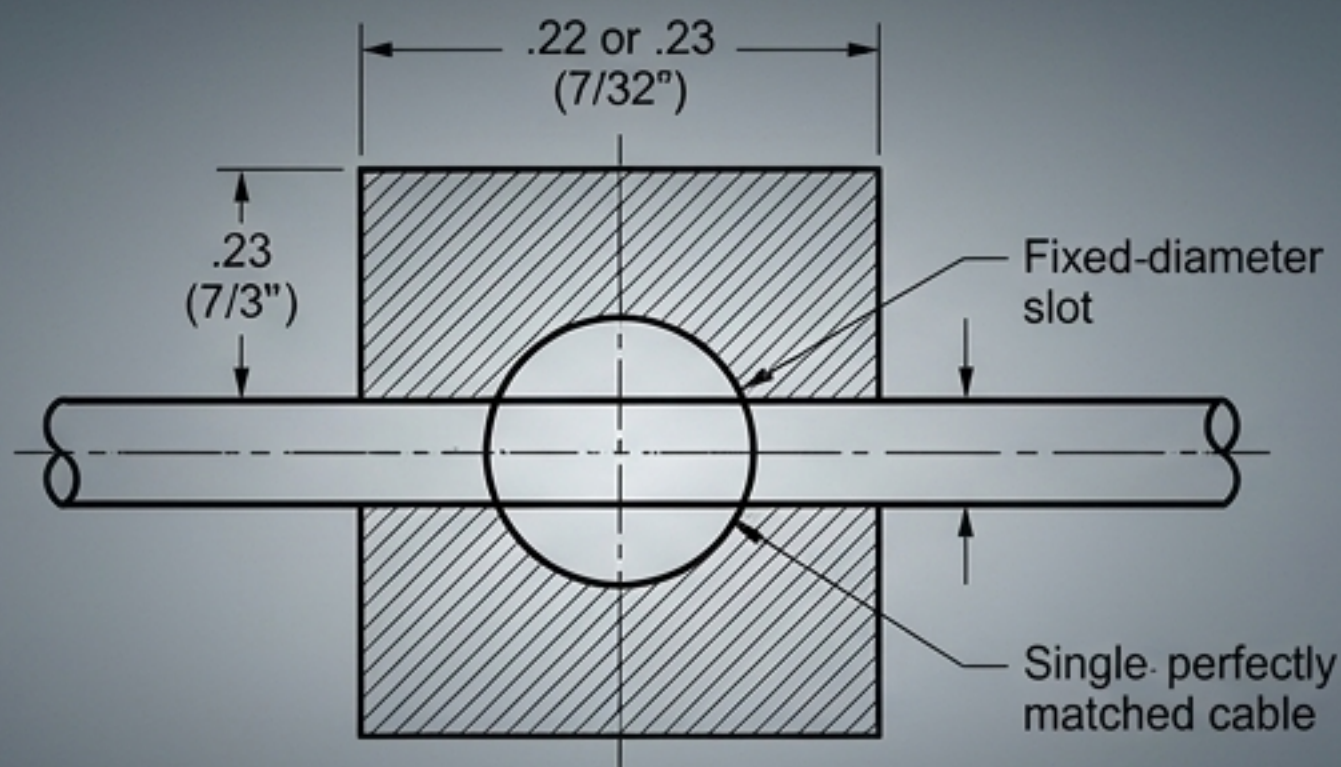
**The Cutter Bar mechanism relies
entirely on a pristine target.**

**It ONLY works
“If the rope socket is clean.”**

**Debris, scale, or unscraped paraffin
accumulation renders the ultimate
backup tool completely useless.**

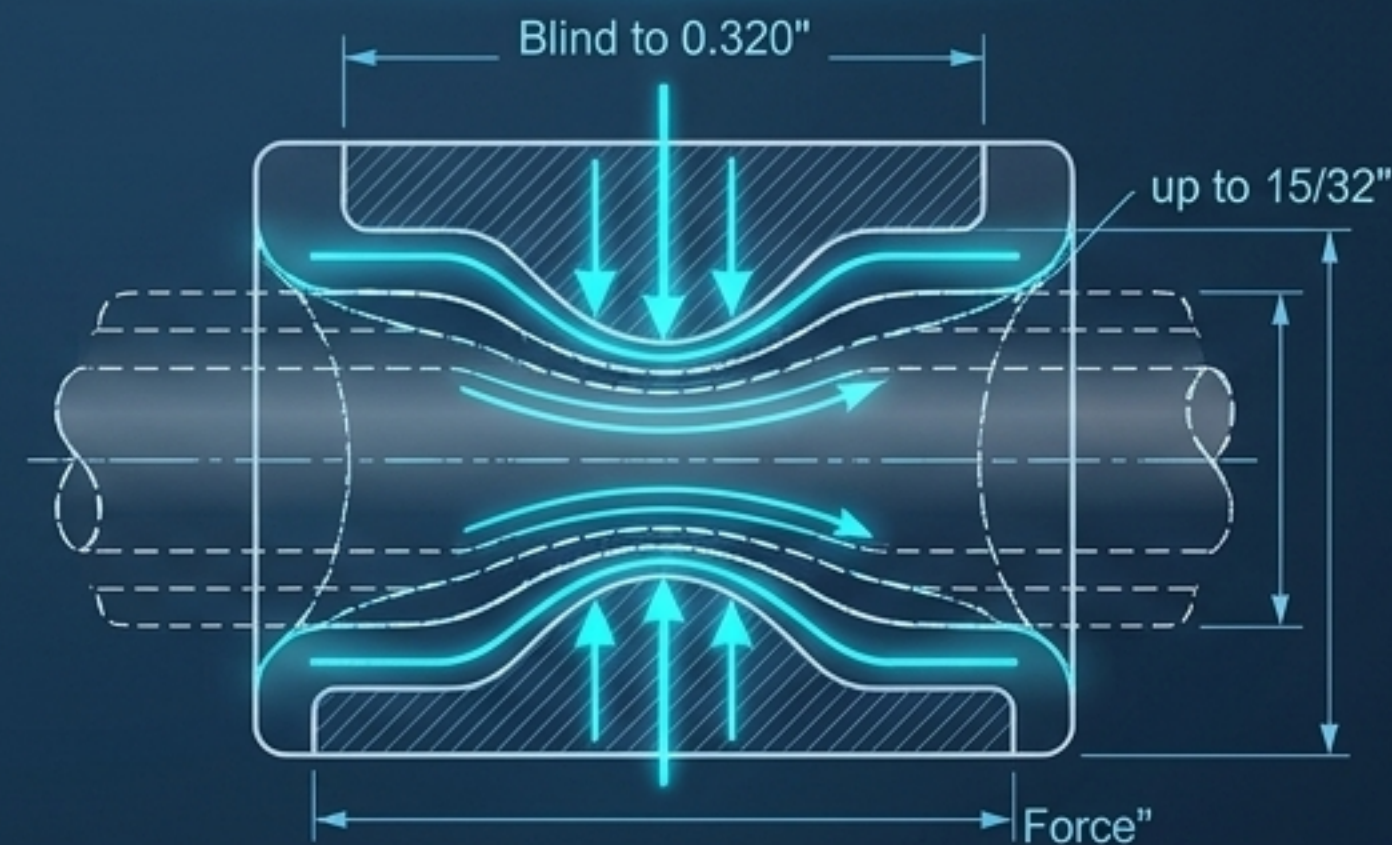
Breaking the Rules of Sizing: The Ram Seal Evolution

The Old Assumption (Conventional Seals)



Conventional inner seals must be slotted perfectly to match exact cable sizes (e.g., .22 or .23 using 7/32"). They are strictly sized for one line only.

The New Reality (Constrictor Multiline Ram Seal)



- **Dynamic Range:** Seals from blind through to 0.320" (and in development up to 15/32").
- **The Game-Changer:** In an unexpected cut and drop, it seals blind, holds fluids for 12 hours, and closes on any cable up to 5/16" without shearing.
- **The Result:** Crews can perform remedial work on the intact cable, a process previously impossible under pressure.

The Blueprint of Mastery: Total System Awareness



Synthesis Statement: Mastery in wireline operations requires abandoning the view of a cable merely as a conveyance tool. It is a live nervous system demanding total awareness.

1. Respect the Funnel

Mechanical ground-truth (LIB/Blind Box) must precede digital and E-line interventions.

2. Doubt the Barrier

Assume equipment has caveats (Clean rope sockets, BOP shear limits). Engineer the operation to never rely solely on a tertiary barrier.

3. Control the Physics

De-pressurization is not downtime; it is a critical thermodynamic and electrostatic operation. Patience is the ultimate safety device.